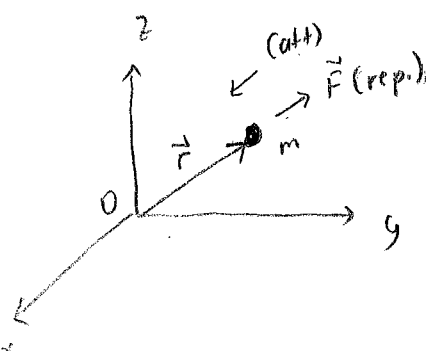


Central Force Motion

A force is central if:

- It must be towards or away from a fixed point (center of force)
- Magnitude of the force should only be proportional to the distance r b/w the particle and the center of force.



$$\vec{F} = f(r) \hat{r}_1 \quad \text{unit vector in the direction of } \vec{r}.$$

- It is an important problem
 - It is completely solvable.
 - Explains the motion of celestial bodies.
 - Explains certain 2 body nuclear interactions (e.g. scattering of α particles from a nuclei)

Ex:

- $\vec{F} = -\frac{3}{r^2} \hat{r}_1$ Att.
- $\vec{F} = 4r^2 \hat{r}_1$ Rep.
- $F = r(r-2) \hat{r}_1$ $r > 2$ rep. $0 < r < 2$ att.

Which one is attractive and which one is repulsive?

Properties of Central Force:

- Central force cannot apply any torque on the particle. So, the angular momentum is conserved.

the same direction

$$\vec{N} = \vec{r} \times \vec{F} = \vec{r} \times [f(r) \hat{r}_1] = f(r) [\vec{r} \times \hat{r}_1] = 0$$

$$\vec{N} = \frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L} = \text{constant}$$

because of this reason we use polar coordinates in central-force motion

- The motion of the particle takes place in a plane:

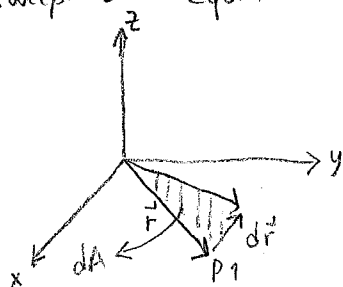
always

$$\vec{L} = \vec{r} \times \vec{p} = m(\underbrace{\vec{r} \times \vec{v}}_{\vec{h}}) = \text{constant} \Rightarrow \vec{r} \text{ and } \vec{v} \text{ must lie in a plane normal to the angular momentum vector } \vec{L}.$$

If $\vec{L}$ is constant, so $\vec{r}$ and $\vec{v}$ will also stay on the same plane

- The position vector $\vec{r}$ of the particle with respect to the center of force sweeps out equal areas in equal times, i.e., areal velocity is constant.

$$dA = \frac{1}{2} |\vec{r} \times d\vec{r}| = \frac{1}{2} |\vec{r} \times \vec{v} dt| \Rightarrow \frac{dA}{dt} = \frac{1}{2} |\vec{r} \times \vec{v}| = \frac{h}{2m}$$



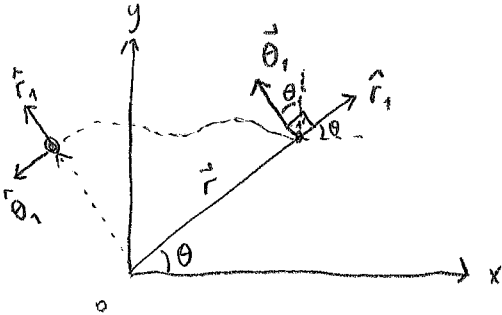
$$\rightarrow A = \frac{h}{2} t + C \rightarrow \text{linear in time}$$

⇒ equal areas in equal time intervals.

Equations of Motion in a Central Force Field

Polar coord. → The most convenient coordinate system because central-force is in the r direction.

How to describe $\vec{r}$, $\vec{v}$ and $\vec{a}$ in polar coordinates?



$\hat{r}_1 = \cos\theta \hat{i} + \sin\theta \hat{j}$
 $\hat{\theta}_1 = -\sin\theta \hat{i} + \cos\theta \hat{j}$ } Unlike unit vectors in cart. coord the polar coord. unit vectors are not fixed in direction. Their direction changes as the particle moves along the path.

$$\frac{d\hat{r}_1}{dt} = -\sin\theta \frac{d\theta}{dt} \hat{i} + \cos\theta \frac{d\theta}{dt} \hat{j} = \frac{d\theta}{dt} \{-\sin\theta \hat{i} + \cos\theta \hat{j}\} = \dot{\theta} \hat{\theta}_1$$

$$\frac{d\hat{\theta}_1}{dt} = -\cos\theta \frac{d\theta}{dt} \hat{i} - \sin\theta \frac{d\theta}{dt} \hat{j} = -\frac{d\theta}{dt} \{\cos\theta \hat{i} + \sin\theta \hat{j}\} = -\dot{\theta} \hat{r}_1$$

In summary: $\frac{d\hat{r}_1}{dt} = \dot{\theta} \hat{\theta}_1$ and $\frac{d\hat{\theta}_1}{dt} = -\dot{\theta} \hat{r}_1$ → Consider the geometrical interpretation too.

Then the velocity of the particle becomes:

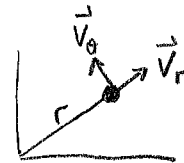
$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r \hat{r}_1) = \frac{dr}{dt} \hat{r}_1 + r \frac{d\hat{r}_1}{dt} = \dot{r} \hat{r}_1 + r \dot{\theta} \hat{\theta}_1$$



we may write $\vec{v} = \dot{r} \hat{r}_1 + r \dot{\theta} \hat{\theta}_1$ as $\vec{v} = \underbrace{(\dot{r}) \hat{r}_1}_{\text{radial velocity}} + \underbrace{(r \dot{\theta}) \hat{\theta}_1}_{\text{tangential velocity}}$

$$V_r = \dot{r}$$

$$V_\theta = r \dot{\theta} = \frac{d(\text{arc length})}{dt}$$



$$v = \sqrt{V_r^2 + V_\theta^2}$$

The acceleration:

$$\frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r} \hat{r}_1 + r \dot{\theta} \hat{\theta}_1) = \ddot{r} \hat{r}_1 + \dot{r} \frac{d\hat{r}_1}{dt} + \dot{r} \dot{\theta} \hat{\theta}_1 + r \ddot{\theta} \hat{\theta}_1 + r \dot{\theta} \frac{d\hat{\theta}_1}{dt}$$

$$= \ddot{r} \hat{r}_1 + \dot{r} \dot{\theta} \hat{\theta}_1 + \dot{r} \dot{\theta} \hat{\theta}_1 + r \ddot{\theta} \hat{\theta}_1 - r \dot{\theta}^2 \hat{r}_1$$

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r}_1 + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{\theta}_1 \xrightarrow{\text{can be written as}} \vec{a} = a_r \hat{r}_1 + a_\theta \hat{\theta}_1 \text{ where } a_r = \ddot{r} - r \dot{\theta}^2$$

$$a_\theta = r \ddot{\theta} + 2\dot{r} \dot{\theta}$$

au. in r direction

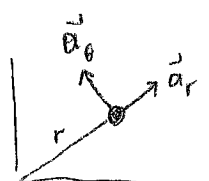
$$\frac{1}{r} \left(\frac{ds}{dt} \right)^2 = \frac{1}{r} \left(r \dot{\theta} \right)^2$$

$$= r \left(\frac{d\theta}{dt} \right)^2 = r \dot{\theta}^2 = r \ddot{\theta}$$

au. in θ dir.

$$\frac{d}{dt} \frac{ds}{dt} = \frac{d}{dt} \frac{d}{dt} (r \dot{\theta})$$

and $a = \sqrt{a_r^2 + a_\theta^2}$



Example: If a particle moves in a plane acc. to expressions $\theta = 0.3t + 0.2t^2$ $r = 0.5t + 0.4t^2$, Find its velocity and acceleration at $t=2$.

$$\dot{r} = 0.5 + 0.8t$$

$$\ddot{r} = 0.8$$

$$V_r(2) = 0.5 + 1.6 = 2.1 \text{ m/s}$$

$$\dot{\theta} = 0.3 + 0.4t$$

$$\ddot{\theta} = 0.4$$

$$V_\theta(2) = r\dot{\theta} = (1 + 1.6)(0.3 + 0.8) = 2.6 \times 1.1 = 2.9 \text{ m/s}$$

$$V = \sqrt{V_r^2 + V_\theta^2} = 3.6 \text{ m/s}$$

! This part will be replaced with 36° !

$$a_r = \ddot{r} - r\dot{\theta}^2 = -2.35 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 5.7 \text{ m/s}^2$$

$$a = \sqrt{a_r^2 + a_\theta^2} = 6.2 \text{ m/s}^2$$

Let us first express angular momentum in polar coordinates: $\vec{L} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$

$$= m(r^2\dot{\theta}\hat{r}_1 \times \hat{\theta}_1)$$

$$= m\vec{h} \rightarrow \text{constant}$$

$$\vec{L} = \underbrace{m r^2 \dot{\theta}}_{\text{magnitude}} \underbrace{(\hat{r}_1 \times \hat{\theta}_1)}_{\text{direction}}$$

$$L = m r^2 \dot{\theta} : \text{conserved}$$

Now we can consider the equations of motion:

$$\vec{F} = m\vec{a}$$

acc. in polar coord.

$$f(r)\hat{r}_1 = m[(\ddot{r} - r\dot{\theta}^2)\hat{r}_1 + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}_1]$$

$$\Rightarrow \left\{ \begin{array}{l} f(r) = m(\ddot{r} - r\dot{\theta}^2) \\ 0 = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \end{array} \right.$$

we will see that not all of these quantities change in time!!!
* we will use the first eq. to come up with a diff. eq. containing r and θ only, so that when we solve it we will have the trajectory of the particle.

We know that $\vec{r} \times \vec{v} = \vec{h}$ is a constant

$$\text{Also velocity in polar coordinates: } \vec{v} = \dot{r}\hat{r}_1 + r\dot{\theta}\hat{\theta}_1$$

$$\Rightarrow \vec{r} \times \vec{v} = r\hat{r}_1 \times (\dot{r}\hat{r}_1 + r\dot{\theta}\hat{\theta}_1) = r\dot{r}\hat{r}_1 \times \hat{r}_1 + r^2\dot{\theta}\hat{r}_1 \times \hat{\theta}_1 = \vec{0} + r^2\dot{\theta}\hat{r}_1 \times \hat{\theta}_1 = \vec{h} = \text{constant}$$

If we assume x-y is the plane of motion, then $\hat{r}_1 \times \hat{\theta}_1$ is parallel to $\hat{k}$ (z-direction)

so we can simply write

$$\vec{h} = \underbrace{r^2\dot{\theta}}_h \underbrace{(\hat{r}_1 \times \hat{\theta}_1)}_{\hat{k}} = r^2\dot{\theta}\hat{k} = h\hat{k} \rightarrow h = r^2\dot{\theta}$$

Therefore: $f(r) = m(\ddot{r} - r\dot{\theta}^2) \rightarrow f(r) = m(\ddot{r} - \frac{h^2}{r^3})$ since $h^2 = r^4\dot{\theta}^2$

• For some calculations making the following change of variables is more suitable:

$$u = \frac{1}{r} \rightarrow r = \frac{1}{u} \rightarrow \dot{r} = -\frac{1}{u^2}\dot{u} \quad \text{also} \quad u^2 = \frac{1}{r^2} \quad \text{and} \quad h = r^2\dot{\theta} \Rightarrow u^2 = \frac{\dot{\theta}}{h}$$

$$\Rightarrow u^2 = \frac{\dot{\theta}}{h} \Rightarrow \dot{r} = -\frac{h}{\dot{\theta}}\dot{u} = -h\left(\frac{du/dt}{d\theta/dt}\right) = -h\frac{du}{d\theta}$$

$$\dot{\theta} = hu^2$$

$$\ddot{r} = \frac{d}{dt}\left(-h\frac{du}{d\theta}\right) = \frac{d}{d\theta}\frac{d\theta}{dt}\left(-h\frac{du}{d\theta}\right) = \frac{d}{d\theta}\left(-h\frac{du}{d\theta}\right)\dot{\theta} = -h\frac{d^2u}{d\theta^2}\dot{\theta} = -h^2u^2\frac{d^2u}{d\theta^2}$$

$$\Rightarrow f(1/u) = m\left\{-h^2u^2\frac{d^2u}{d\theta^2} - h^2u^3\right\}$$

$$\frac{1}{m}f(1/u) + h^2u^3 = -h^2u^2\frac{d^2u}{d\theta^2}$$

$$\frac{d^2u}{d\theta^2} + \frac{h^2u^3}{h^2u^2} = -\frac{1}{mh^2u^2}f(1/u) \rightarrow \frac{d^2u}{d\theta^2} + u = -\frac{1}{mh^2u^2}f(1/u)$$

$$\frac{d^2u}{d\theta^2} + u = -\frac{1}{mh^2u^2}f(1/u)$$

This is the equation of path in central force field.

Central Force Motion (Summary)

$$\vec{F}(r) = f(r) \hat{r}$$

- ① $\vec{N} = \frac{d\vec{L}}{dt} = 0$ Any mom. conserved
- ② Motion takes place on a plane $\rightarrow$ polar coordinates (r, θ)
- ③ Position vector scans equal areas in equal times.

Velocity and acceleration in polar coordinates

$$\vec{V} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \hat{\theta}$$

Equations of motion:

$$f(r) = m(\ddot{r} - r \dot{\theta}^2)$$

$$0 = m(r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \Rightarrow \text{no tangential force.}$$

Expressing $\vec{L}$ in polar coordinates

$$\vec{L} = m r^2 \dot{\theta} (\hat{r} \times \hat{\theta}) \quad |\vec{L}| = L = m r^2 \dot{\theta}$$

Expression of $\vec{L}$ in polar coordinates

$$\vec{L} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v}) = m[(r\hat{r}_1) \times (\dot{r}\hat{r}_1 + r\dot{\theta}\hat{\theta}_1)] = m[\underbrace{r\dot{r}(\hat{r}_1 \times \hat{r}_1)}_0 + r^2\dot{\theta}(\hat{r}_1 \times \hat{\theta}_1)]$$

$$\vec{L} = \underbrace{m r^2 \dot{\theta}}_{\text{mag. dir.}} (\hat{r}_1 \times \hat{\theta}_1) \Rightarrow |\vec{L}| = l = m r^2 \dot{\theta} \xrightarrow[\text{of } h]{\text{or in terms of } h} l = m h : \text{constant unnecessary}$$

Equations of motion for central force problem

$$\vec{F} = m\vec{a}$$

$$f(r)\hat{r}_1 = m[(\ddot{r} - r\dot{\theta}^2)\hat{r}_1 + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}_1]$$

$$\Rightarrow \boxed{f(r) = m(\ddot{r} - r\dot{\theta}^2)} \rightarrow a_{\theta} : 0 = m a_{\theta} \Rightarrow \text{no tangential force is applied on the particle.}$$

$$0 = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) \rightarrow \text{Does this equation a trivial result or does it provide us sth? (look at the back page)}$$

As the next step, all we want to express the first equation in terms of r and θ only because we want to obtain the trajectory of the particle in polar coordinates $r(\theta)$ after solving the differential equation. So,

$$f(r) = m(\ddot{r} - r\dot{\theta}^2)$$

Let's first insert the constant of motion into this equation: (we are getting rid of time dependency)

$$l^2 = m^2 r^4 \dot{\theta}^2 \Rightarrow \dot{\theta}^2 = \frac{l^2}{m^2 r^4} \Rightarrow f(r) = m\left[\ddot{r} - \frac{r l^2}{m^2 r^4}\right] = m\left[\ddot{r} - \frac{l^2}{m^2 r^3}\right]$$

• We want to get rid of the time dependence in this last expression, where it obviously manifests itself in the term $\ddot{r}$.

• For this reason we will consider change of variables: $r = 1/u$

$$\Rightarrow \dot{r} = -\frac{1}{u^2} \dot{u} \quad \text{and} \quad \frac{1}{u^2} = r^2 \quad \text{and} \quad r^2 = l/m\dot{\theta} \Rightarrow \dot{r} = -\frac{l}{m\dot{\theta}} \dot{u}$$

$$\Rightarrow \ddot{r} = -\frac{l}{m} \left[\frac{d\dot{u}/dt}{d\theta/dt} \right] = -\frac{l}{m} \frac{du}{d\theta}$$

to get rid of the time dependence $\frac{l}{mr^2} = \frac{lu^2}{m}$

$$\ddot{r} = -\frac{l}{m} \frac{d}{dt} \left(\frac{du}{d\theta} \right) = -\frac{l}{m} \frac{d}{d\theta} \frac{d\theta}{dt} \left(\frac{du}{d\theta} \right) = -\frac{l}{m} \frac{d^2 u}{d\theta^2} \dot{\theta} = -\frac{l}{m} \frac{lu^2}{m} \frac{d^2 u}{d\theta^2} = -\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2}$$

$$= f(1/u) = m \left[-\frac{l^2 u^2}{m^2} \frac{d^2 u}{d\theta^2} - \frac{l^2}{m^2} u^3 \right] = -\frac{l^2}{m} \left[u^2 \frac{d^2 u}{d\theta^2} + u^3 \right]$$

$$\Rightarrow -\frac{m}{l^2} f\left(\frac{1}{u}\right) = u^2 \frac{d^2 u}{d\theta^2} + u^3 \Rightarrow \boxed{-\frac{m}{u l^2} f\left(\frac{1}{u}\right) = \frac{d^2 u}{d\theta^2} + u} \xrightarrow{\text{also}} \boxed{\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{m}{l^2} r^2 f(r)}$$

$\vec{h}$ was defined as $\vec{h} \equiv \vec{r} \times \vec{v}$. Sometimes it can be more convenient to consider the angular momentum $\vec{L} \equiv m(\vec{r} \times \vec{v})$ and its magnitude $m|\vec{r} \times \vec{v}| = \ell \Rightarrow \boxed{\ell = m h}$

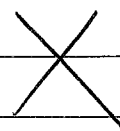
* We used the quantity $h = |\vec{r} \times \vec{v}|$ not to involve m in our above calculations.

So, equation of path can also be expressed in terms of angular momentum;

$$\frac{d^2 u}{d\theta^2} + u = -\frac{1}{m \frac{\ell^2}{m^2} u^2} f(1/u) \Rightarrow \boxed{\frac{d^2 u}{d\theta^2} + u = -\frac{m}{\ell^2 u^2} f(1/u)}$$

This equation can also be written as:

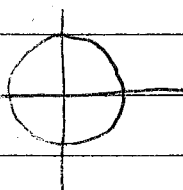
$$\boxed{\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{m}{\ell^2} r^2 f(r)}$$



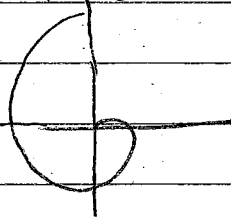
Example: Find the force law for central-force field that allows a particle to move in a logarithmic spiral orbit given by $r = k e^{\alpha\theta}$ where k and α are constants.

→ First of all $r(\theta)$ defines a polar function r changes with respect to θ .

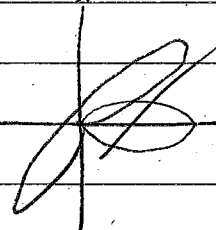
For ex: $r=1$



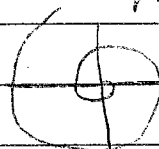
$r=\theta$



$r = \sin \theta$



$r = k e^{\alpha\theta}$



$$\begin{aligned} \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) &= \frac{d^2}{d\theta^2} \left(\frac{1}{k e^{\alpha\theta}} \right) = \frac{1}{k} \frac{d^2}{d\theta^2} (e^{-\alpha\theta}) = \frac{-\alpha}{k} \frac{d}{d\theta} (e^{-\alpha\theta}) = \frac{\alpha^2}{k} e^{-\alpha\theta} \\ &= \frac{\alpha^2}{k e^{\alpha\theta}} = \frac{\alpha^2}{r} \end{aligned}$$

$$\Rightarrow f(r) = - \left(\frac{\alpha^2}{r} + \frac{1}{r} \right) \frac{\ell^2}{m r^2} = - \frac{\ell^2 (\alpha^2 + 1)}{m r^3} \Rightarrow \text{Attractive inverse cube}$$

Example: Determine $r(t)$ and $\theta(t)$ for the above problem.

We know that $\ell = m r^2 \dot{\theta}$

$$\dot{\theta} = \frac{\ell}{m r^2} = \frac{\ell}{m k^2 e^{2\alpha\theta}} \rightarrow e^{2\alpha\theta} d\theta = \frac{\ell}{m k^2} dt \rightarrow \text{Integrate both sides}$$

$$\frac{1}{2\alpha} e^{2\alpha\theta} = \frac{\ell t}{m k^2} + C' \quad \xrightarrow{\text{multiply with } 2\alpha} \quad e^{2\alpha\theta} = \frac{2\alpha \ell t}{m k^2} + (2\alpha C')$$

$$\ln e^{2\alpha\theta} = \ln \left[\frac{2\alpha \ell t}{m k^2} + C \right] \rightarrow \theta(t) = \frac{1}{2\alpha} \ln \left(\frac{2\alpha \ell t}{m k^2} + C \right)$$

We also know that $\frac{\ell}{mr^2} = \frac{\ell}{mk^2 e^{2\alpha\theta}} \Rightarrow \frac{k^2}{r^2} = \frac{1}{e^{2\alpha\theta}} \Rightarrow \frac{r^2}{k^2} = e^{2\alpha\theta}$

$\Rightarrow \frac{r^2}{k^2} = \frac{2\alpha\ell t}{mk^2} + C \Rightarrow r(t) = \left[\frac{2\alpha\ell}{m} t + k^2 C \right]^{1/2}$ C and ℓ are determined from the initial conditions

The Total energy in central force problem:

If the force is conservative and the potential of the system does not depend on time then we know that the total energy is conserved.

$T + U = E \rightarrow$ constant for the central force problem.

$T = \frac{1}{2} m v^2 = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) \Rightarrow E = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + U(r)$
 $\frac{1}{2} I \omega^2$: rotational energy $\ell = m r^2 \dot{\theta}$

or $E = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 + U(r) = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} \frac{\ell^2}{m r^2} + U(r)$

since $\ell = m r^2 \dot{\theta}$ at this stage we start considering the central force. $= \frac{1}{2} m \dot{r}^2 + \frac{1}{2} \frac{\ell^2}{m r^2} + U(r)$

$\rightarrow$ when $U(r)$ is specified $E = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} \frac{\ell^2}{m r^2} + U(r)$ completely describes the system.

Alternative to the previous equation of motion, E can be integrated to obtain the general solution in terms of E and ℓ .

$\Rightarrow 2(E - U) = m \dot{r}^2 + \frac{\ell^2}{m r^2} \rightarrow \frac{2}{m} (E - U) = \dot{r}^2 + \frac{\ell^2}{m^2 r^2}$

$\rightarrow \dot{r}^2 = \frac{2}{m} (E - U) - \frac{\ell^2}{m^2 r^2} \rightarrow \dot{r} = \pm \sqrt{\frac{2}{m} (E - U) - \frac{\ell^2}{m^2 r^2}} = \frac{dr}{dt}$

This eq. can be solved for dt and integrated to obtain $t = t(r)$

An inversion of this result would yield $r = r(t)$. However, we are more interested in equation of the path in terms of r and θ . Therefore,

$d\theta = \frac{d\theta}{dt} \frac{dt}{dr} dr = \frac{\dot{\theta}}{\dot{r}} dr$

$\ell = m r^2 \dot{\theta} \rightarrow \dot{\theta} = \ell / m r^2$

$d\theta = \frac{\pm \ell / m r^2}{\sqrt{\frac{2}{m} (E - U) - \frac{\ell^2}{m^2 r^2}}} dr = \frac{\pm (\ell / r^2) dr}{\sqrt{2m(E - U) - \frac{\ell^2}{r^2}}} = \frac{\pm (\ell / r^2) dr}{\sqrt{2m \left[E - U - \frac{\ell^2}{2mr^2} \right]}}$

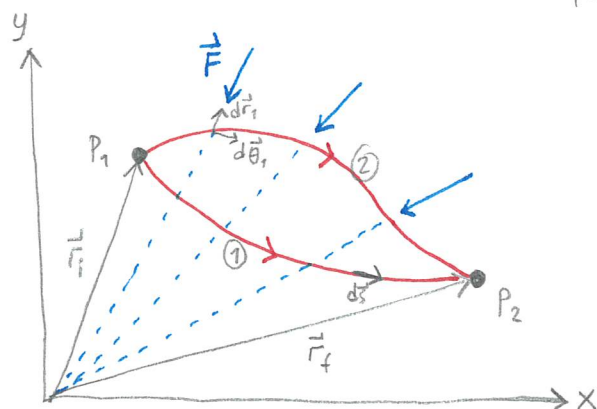
$\theta(r) = \int \frac{\pm \ell / r^2}{\sqrt{2m \left[E - U - \frac{\ell^2}{2mr^2} \right]}} dr$

IMPORTANT OBSERVATION:

ℓ is constant in time, $\dot{\theta}$ cannot change sign, so $\theta(t)$ must either increase or decrease monotonically. (38)

In the beginning of "The total energy in central force problem":

(Why central force is conservative; independent of path?)



A particle is moving from P_1 to P_2 while a central force located at the origin acting on it. The paths (1) and (2) are combinations of radial and curved segments. Central force always acts in the direction of radial segments. It is perpendicular to the displacement along any curved segments.

$$d\vec{r}_1 \parallel \vec{F} \text{ and } d\vec{\theta}_1 \perp \vec{F}$$

Thus, the work done by the central force along any curved segment is zero, and the total work done in moving the particle along any path is equal to the work done along a radial line from $\vec{r}_i$ to $\vec{r}_f$. That is, the work done by a central force is independent of path.

$$W = \int_1^2 \vec{F} \cdot d\vec{s} \quad \rightarrow \quad d\vec{s} = dx\hat{i} + dy\hat{j} \quad d\vec{s} = dr\hat{r}_1 + d\theta\hat{\theta}_1$$

Cartesian coord. polar coordinate.

$$\Rightarrow \vec{F} \cdot d\vec{s} = f(r)\hat{r}_1 \cdot (dr\hat{r}_1 + d\theta\hat{\theta}_1) = f(r)dr\hat{r}_1 \cdot \hat{r}_1 + f(r)d\theta\hat{r}_1 \cdot \hat{\theta}_1$$

$$= f(r)dr$$

$$W = \int_{r_i}^{r_f} f(r)dr$$

The work done by the central force only depends on r_i and r_f , no dependence on the path in between.

★ From the discussion above, we conclude that "CENTRAL FORCE IS CONSERVATIVE". So there exists a potential U

About $\Theta(r)$ equation:

It can only be solved if $F(r) \propto r^n$ for $n=1, -2, -3$

In these cases solutions are expressible in terms of sines and cosines.

- $n=1$ is the harmonic oscillator problem

$$F(r) \propto r \rightarrow F(r) = -kr$$

- $n=-2$: inverse-square-law : we will solve this

They have prime importance in physical situations.

Example: What is the total energy of "the orbit" in the previous two examples?

$$F = - \frac{dU}{dr} \quad ; \quad \text{Conservative Force} \rightarrow U = - \int F dr$$

we have found that:

$$F(r) = - \frac{\ell^2}{mr^3} (\alpha^2 + 1)$$

attractive.

$$U = + \int \frac{\ell^2}{mr^3} (\alpha^2 + 1) dr = \frac{\ell^2 (\alpha^2 + 1)}{m} \int r^{-3} dr = - \frac{\ell^2 (\alpha^2 + 1)}{2mr^2} \quad U=0 \text{ as } r \rightarrow \infty$$

$$E = \underbrace{\frac{1}{2} m \dot{r}^2}_{\text{rel. to radial vel.}} + \underbrace{\frac{1}{2} m r^2 \dot{\theta}^2}_{\text{rel. to tangential vel.}} + U(r)$$

$$= \frac{1}{2} m \dot{r}^2 + \frac{1}{2} \frac{\ell^2}{mr^2} + U(r)$$

★ This is an important expression, because it is actually the orbit condition. We internally assume that r is connected with θ , i.e., there is an orbit. However, if r is completely independent of θ , we can set $\frac{1}{2} m r^2 \dot{\theta}^2 \rightarrow 0$ and total energy will not be zero.

we are also given:

$$r = k e^{\alpha \theta}$$

$$\frac{dr}{d\theta} = k \alpha e^{\alpha \theta}$$

To find $\dot{r}$ we can use $\ell = m r^2 \dot{\theta}$

$$\dot{\theta} = \frac{\ell}{m r^2}$$

$$\dot{r} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \alpha k e^{\alpha \theta} \frac{\ell}{m r^2} = \frac{\alpha \ell}{m r} = \frac{\alpha \ell}{mr}$$

$$\Rightarrow E = \frac{1}{2} m \frac{\alpha^2 \ell^2}{m^2 r^2} + \frac{1}{2} \frac{\ell^2}{mr^2} - \frac{\ell^2 (\alpha^2 + 1)}{2mr^2} = \frac{\alpha^2 \ell^2 + \ell^2}{2mr^2} - \frac{\ell^2 (\alpha^2 + 1)}{2mr^2} = 0$$

★ The total energy of the orbit is 0 if $U(r \rightarrow \infty) = 0$, why? Assume that the particle was at infinity without any kinetic energy. Then the total energy of the system would be 0. This must be conserved whatever happens. Consider the work done by the central force to bring the particle from infinity to a point r' . $\vec{F}_c$ and $d\vec{r}$ are in the same direction $\vec{F} \cdot d\vec{r} > 0$

$$W = \int_{\infty}^{r'} \vec{F} \cdot d\vec{r} = \int_{\infty}^{r'} \frac{\ell^2}{mr^3} (\alpha^2 + 1) dr = - \frac{\ell^2 (\alpha^2 + 1)}{2m} \frac{1}{r^2} \Big|_{\infty}^{r'} = - \frac{\ell^2 (\alpha^2 + 1)}{2m r'^2} - 0 < 0$$

→ It is reasonable since F_c is the only force acting on this particle. If it accelerate the particle up to position r' , so the particle will have some positive K.E. in the amount of $+\frac{\ell^2 (\alpha^2 + 1)}{2m r'^2}$ since the potential and kinetic energies must add up to zero (The particle had 0 total energy in the beginning.)

Irrelevant

★ If an external agent brings the particle from ∞ to a distance r ; then the work done by this external force would be

$$W_e = \frac{l^2(\alpha^2 + 1)}{2mr^2} > 0$$

without

→ This is reasonable because to bring the particle from infinity to point r , the external agent would require to apply r .

★ The total energy of the orbit is zero since we assumed there is a relation between r and θ in the form $r = r(\theta)$ so, we were able to write total energy

$$E = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2 + U(r) \rightarrow \dot{\theta} \neq 0 \text{ and } r \text{ does not change independent from } \theta \text{ so that we can write}$$

$$\dot{r} = \frac{dr}{d\theta} \frac{d\theta}{dt} \text{ which results in } E = 0$$

★ Logarithmic spiral orbits $r = k e^{\alpha \theta}$ require the total energy to be zero!

Orbits in central field:

Remember that : $E = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} \frac{l^2}{m r^2} + U(r)$

$$\dot{r} = \frac{dr}{dt} = \pm \sqrt{\frac{2}{m} (E - U) - \frac{l^2}{m^2 r^2}}$$

Let consider the r values where $\dot{r}$ vanishes:

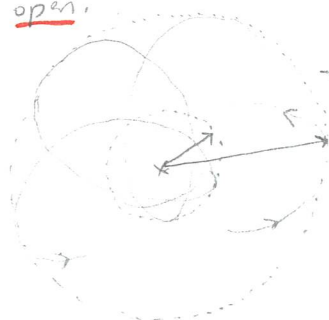
$$\pm \sqrt{\frac{2}{m} (E - U) - \frac{l^2}{m^2 r^2}} = 0 \rightarrow \frac{2}{m} (E - U) - \frac{l^2}{m^2 r^2} = 0$$

$$\rightarrow E - U(r) - \frac{l^2}{2mr^2} = 0$$

★ $\dot{r} = 0$ implies a turning point in the motion. Since the equation is second degree there are 2 roots of this equation $r_{\max}$ and $r_{\min}$, in general. Therefore, the motion takes place in the region $r_{\min} \leq r \leq r_{\max}$.

★ Certain combination of $U(r)$, E and l may produce only a single root for this equation, $\dot{r} = 0$ i.e. $r = \text{constant}$ and the orbit is circular.

★ If the motion is periodic between $r_{\min} \leq r \leq r_{\max}$ then the orbit is closed. If the orbit does not close on itself after some number of oscillations, then it is open.



→ This figure is not drawn correctly!
; see the one on the back side.

We can decide by calculating the change in the angle θ as r goes from $r_{\min}$ to $r_{\max}$ and to again $r_{\min}$.

$$\Delta\theta = 2 \int_{r_{\min}}^{r_{\max}} \frac{(l/r^2) dr}{\sqrt{2m(E - U - \frac{l^2}{2mr^2})}}$$

→ Since the motion is symmetric in time.

The path is closed only if $\Delta\theta$ is a rational fraction of 2π

That is:

• $\Delta\theta = 2\pi \frac{a}{b}$, where a and b are integers.



(Problem 8.35)

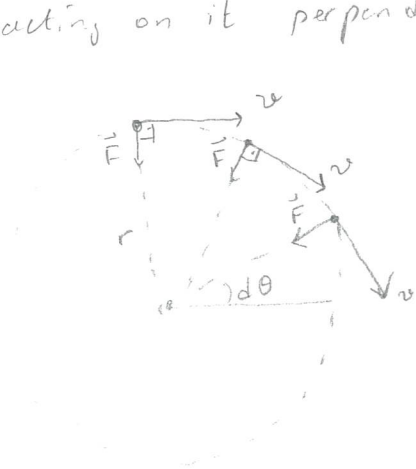
★ After b periods the radius vector returns to its original position.

★ where? we will see in an example that a non-circular closed path can only exist for $U(r) \propto r^{n+2}$ where $n = -2$ and $n = 1$.

- $n = -2$: inverse square law
- $n = 1$: harmonic oscillator.

Circular motion in a nutshell (This will be used to basic concepts)

A particle does a circular motion if there is always a constant force acting on it perpendicular to its direction of motion:



angular velocity

$$\omega = \frac{d\theta}{dt} = 2\pi f = \frac{2\pi}{T}$$

frequency

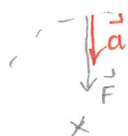
Linear speed:

$$v = \frac{2\pi r}{T} = \omega r$$

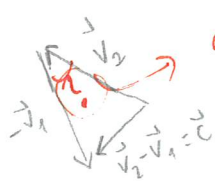
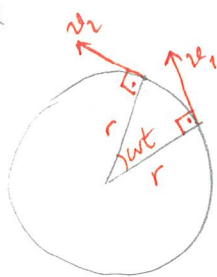
Angular acceleration:

$$\alpha = \frac{d\omega}{dt} = 0 \text{ for a uniform circular motion.}$$

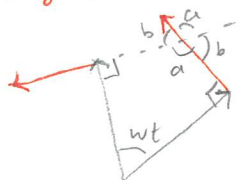
★ Since the only force acting on the particle is the "centripetal force" (the force always showing the centre of the circular path) the only acceleration of the particle always shows the centre of the circle according to Newton's 2nd law.



How to calculate this acceleration? we know that acceleration is time rate of change of velocity. i.e., $|\vec{v}|$ is constant but $\vec{v}$ changes direction.



what is this angle?



Interior angles of a quadrilateral

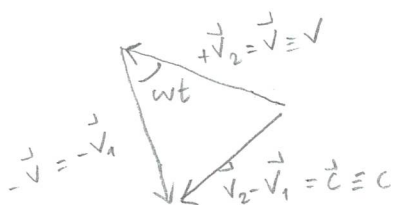
$$90 + 90 + a + wt = 360$$

$$a = 180 - wt$$

$$2/(180 - wt) + 2b = 360$$

$$180 - wt + b = 180$$

$$b = wt$$



$$c = v \omega t$$

$$= \omega t \omega r$$

$$= \omega^2 t r$$

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{change in time}}$$

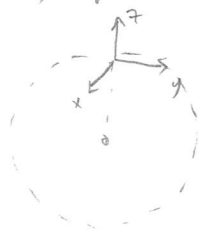
$$= \frac{\omega^2 r t}{t} = \omega^2 r$$

$$v = \omega r$$

$$a_c = \omega^2 r = \frac{v^2}{r^2} r = \frac{v^2}{r}$$

$$\text{Centripetal force} : F = ma = m \frac{v^2}{r} = \frac{m \omega^2 r^2}{r} = m \omega^2 r$$

★ The only real force in circular motion is the "centripetal force". However, sometimes it is useful to think about "centrifugal force". Centrifugal force is only defined when we attach our reference frame on the rotating particle.



It is a non-inertial reference frame since Newton's 1st law is not applicable. (Think about a car taking a turn; we experience a "force" directed opposite to the center of rotation; it is the centrifugal force. It is not a real force, but it is still useful to describe some phenomena like rotation of tornadoes, or why water rotates around a sink while it is being drained).

★ Centripetal and centrifugal forces have the same magnitude $m\omega^2 r$.
Centrifugal energy and effective potential in central force problem
 While calculating $\ddot{r}$ or $\Delta\theta$ ★ See the motivation sheet, page 1

$$\sqrt{E - U - \frac{l^2}{2mr^2}} \quad \text{term is common and crucial.}$$

Especially the term $\frac{l^2}{2mr^2}$ has the units of energy

$$\frac{m^2 \text{ kg}^2 \text{ m}^2 \text{ s}^{-2}}{\text{kg m}^2 \text{ s}^{-2}} = \text{J.}$$

It can be also written by considering $l = mr^2 \dot{\theta}$

$$\frac{l^2}{2mr^2} = \frac{m^2 r^4 \dot{\theta}^2}{2m r^2} = \frac{m r^2 \dot{\theta}^2}{2} = \frac{1}{2} m r^2 \dot{\theta}^2 \rightarrow \text{energy related with tangential velocity.}$$

Exercise 2

$$M_S \approx 2 \times 10^{30} \text{ kg} \quad R_S \approx 700 \times 10^3 \text{ km} = 7 \times 10^5 = 0.7 \times 10^6 \text{ km} \quad \frac{150}{0.7} \approx 200$$

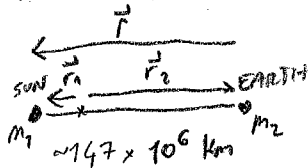
$$M_E \approx 6 \times 10^{24} \text{ kg}$$

$$\text{Dist earth-sun} = 150 \times 10^6 \text{ km}$$

Consider that earth is the only planet rotating around sun. Calculate how far the center of mass is in this 2 body problem from the center of the sun?

$$\vec{r}_1 = \frac{m_2}{m_1 + m_2} \vec{r}$$

$$\vec{r}_2 = \frac{m_1}{m_1 + m_2} \vec{r}$$



$$|\vec{r}_1| = \frac{m_2}{m_1 + m_2} |\vec{r}|$$

$$|\vec{r}_2| = \frac{m_1}{m_1 + m_2} |\vec{r}|$$

$$|\vec{r}_1| = \frac{6 \times 10^{24} \text{ kg}}{(2 \times 10^{30} + 6 \times 10^{24}) \text{ kg}} \times 700 \times 10^3 \text{ km} = \frac{6 \times 10^{24} \text{ kg}}{2 \times 10^{30} \text{ kg}} \times 700 \times 10^3 \text{ km} = 3 \times 10^{-6} \times 700 \times 10^3 \text{ km}$$

$$= 2100 \times 10^{-3} \text{ km} = 2.1 \times 10^{-1} \text{ km} = 2.1 \text{ km}.$$

The length of a football field is 100m. So 2.1 km is approximately 10^5 smaller than the radius of sun. So

$$\frac{0.00100 \text{ m}}{10^5} = 0.001 \text{ m} = 1 \text{ mm}$$

$$\frac{10^6}{10^5} = 1000$$

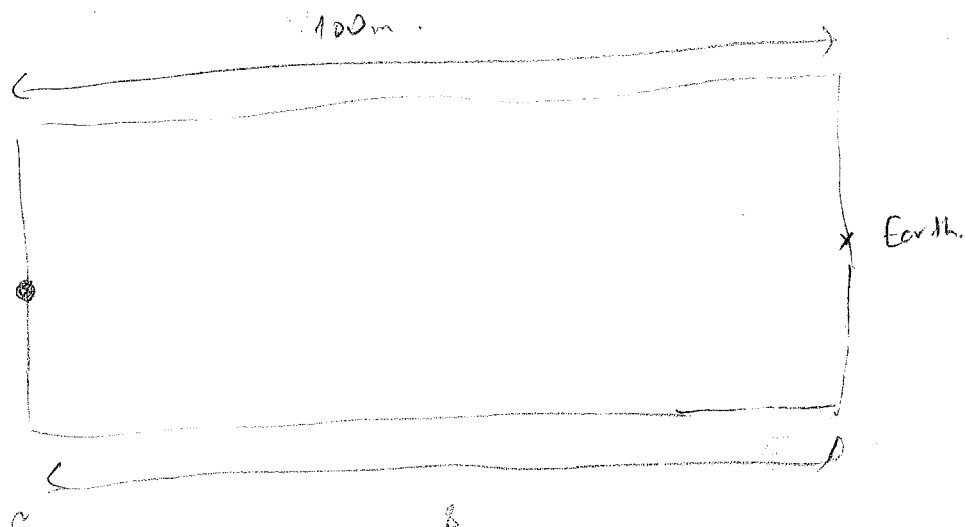
$$150 \times 10^6 \text{ km} = 1.5 \times 10^8 \text{ km}$$

$$R_S = 7 \times 10^5$$

$$0.1 \text{ m} > 10 \text{ cm}$$

$$0.5 \sim$$

$$0.5 \sim$$



Sun's radius
 $7 \times 10^5 \text{ km}$
 $\sim 10^6 \text{ km}$



$$\frac{7 \times 10^5}{2.1} = 3 \times 10^5$$

$$0.000003 \text{ m}$$

$$0.03 \text{ mm}$$

If we interpret this quantity $\frac{1}{2} m r^2 \dot{\theta}^2$ as a "potential energy" $\equiv U_c$, then the "force" associated with U_c can be written as:

$$U_c = \frac{l^2}{2mr^2} \Rightarrow F_c = -\frac{dU}{dr} = -\frac{l^2}{2m} (-2) \frac{1}{r^3} = \frac{l^2}{m} \frac{1}{r^3} = \frac{m l^2 \dot{\theta}^2}{r^3} = m r \dot{\theta}^2$$

This quantity $m r \dot{\theta}^2$ is traditionally called the centrifugal force, and it is not surprising since we have already found out in the previous section that the centrifugal force in a circular motion is $m r \omega^2$.

If we consider the total energy in the central force problem

$$E = \underbrace{\frac{1}{2} m \dot{r}^2}_{\text{energy related to radial vel.}} + \underbrace{\frac{1}{2} m r^2 \dot{\theta}^2}_{\text{energy related to tangential velocity}} + \underbrace{U(r)}_{\text{potential energy.}}$$

$\rightarrow \frac{l^2}{2mr^2}$

? { We actually did nothing but reinterpreted the term $\frac{1}{2} m r^2 \dot{\theta}^2$ as a "potential energy" which we will be calling centrifugal energy.
related to the centrifugal force $m r \dot{\theta}^2$. Why? Because this is an energy related with a motion with a fixed radius. As the particle keep following, for example, a circular path, this quantity will be constant like the real potential energy $U(r)$.

$\rightarrow \frac{1}{2} m r^2 \dot{\theta}^2 \rightarrow \text{constant}$

Therefore, if we interpret $\frac{l^2}{2mr^2}$ as the centrifugal potential energy of the particle, and it can be included with $U(r)$ in an effective potential energy defined by

$$V(r) \equiv U(r) + \frac{l^2}{2mr^2}$$

Therefore the total energy of the central force problem can be written as

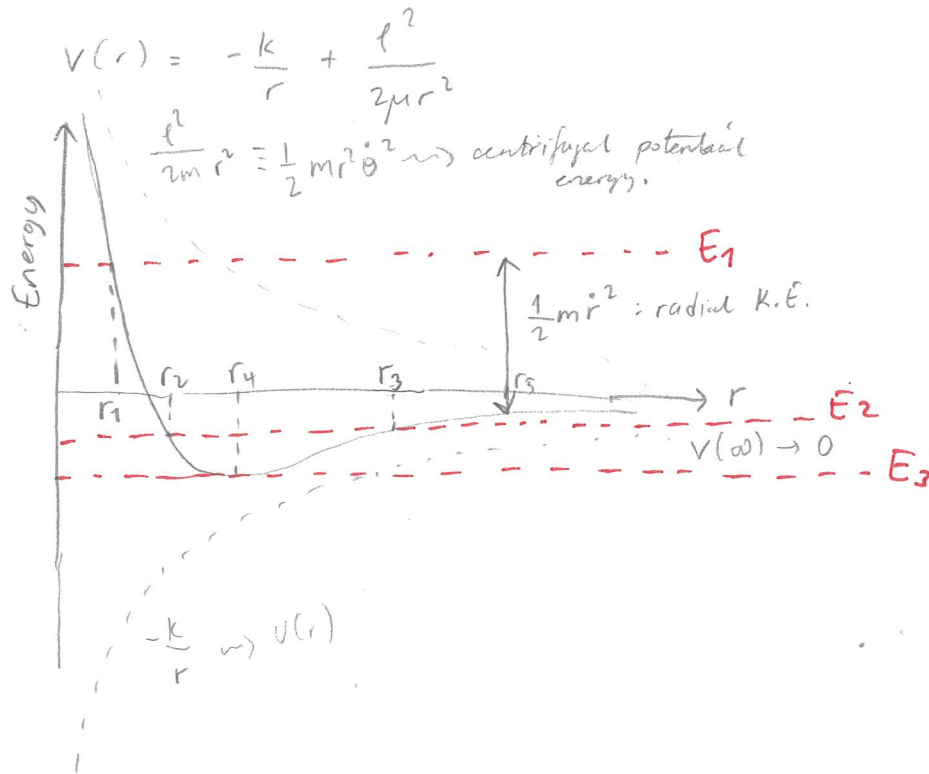
$$E = \frac{1}{2} m \dot{r}^2 + \underbrace{\frac{1}{2} m r^2 \dot{\theta}^2}_{\text{centrifugal potential energy}} + \underbrace{U(r)}_{\text{potential energy}} = \frac{1}{2} m \dot{r}^2 + V(r)$$

effective potential energy.

★ $V(r)$ is a fictitious potential that combines the real potential function $U(r)$ with the energy term associated with the angular motion about the center of force.

For example: If we consider inverse-square-law central-force motion

$$F(r) = -\frac{k}{r^2} \rightarrow U(r) = -\int F(r) dr = -\frac{k}{r}$$



This graph tells us a lot about the dynamics. We have three regions of interest

i) $E = 0$ or $E > 0 \rightarrow$ motion is unbounded (like E_1) for E_1

$\rightarrow$ If the particle moves towards $r=0$, it strikes the potential barrier at r_1 (turning point) and reflected back towards infinitely large r .

Notice that $E_1 - V(r) = \frac{1}{2} m \dot{r}^2$ radial K.E. of the particle

It becomes 0 at r_1 and $\dot{r}$ changes sign after that (when particle hits r_1)

ii) If $E < 0$ and lies between 0 and the min of $V(r)$ (like E_2) then the motion is bounded between $r_2 \leq r \leq r_3$. The values r_2 and r_3 are the turning points.

iii) If E equals the min. of the effective potential (like E_3) then $\dot{r} = 0$ for all values of time and the motion is circular.

$\rightarrow$ Values less than $V_{\min}$ do not result in physically real motion $\dot{r}^2 < 0$ and velocity is imaginary.

• When we consider the total energy of the system:

(*)

$$E = \frac{1}{2} m_1 \dot{\vec{r}}_1^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2^2 + U(r) \quad \text{together with} \quad \boxed{\vec{r}_1 = \frac{m_2}{m_1+m_2} \vec{r}, \quad \vec{r}_2 = \frac{m_1}{m_1+m_2} \vec{r}}$$

$$E = \frac{1}{2} m_1 \left(\frac{m_2}{m_1+m_2} \right)^2 \dot{\vec{r}}^2 + \frac{1}{2} m_2 \left(\frac{m_1}{m_1+m_2} \right)^2 \dot{\vec{r}}^2 + U(r)$$

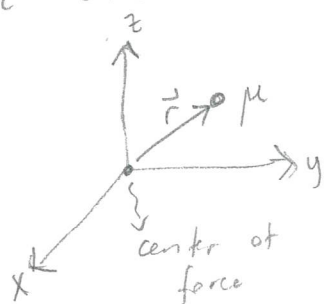
$$= \frac{1}{2} \left\{ m_1 \left(\frac{m_2}{m_1+m_2} \right)^2 + m_2 \left(\frac{m_1}{m_1+m_2} \right)^2 \right\} \dot{\vec{r}}^2 + U(r)$$

$$\frac{m_1 m_2^2 + m_2 m_1^2}{(m_1+m_2)^2} = \frac{m_1 m_2 (m_2 + m_1)}{(m_1+m_2)^2} = \frac{m_1 m_2}{m_1+m_2} \equiv \mu : \text{reduced mass.}$$

$$\Rightarrow E = \frac{1}{2} \mu \dot{\vec{r}}^2 + U(r) \rightarrow \sqrt{\frac{2}{\mu} [E - U(r)]} = \dot{r}$$

$$t - t_0 = \int_{r_0}^r \frac{\pm dr}{\sqrt{\frac{2}{\mu} [E - U(r)]}}$$

$\Rightarrow$ Therefore, we reduced the problem of the motion of two bodies to an equivalent one-body problem with mass μ described by a central-force problem with potential $U(r)$



If we solve this central-force problem with the reduced mass μ and obtain $\vec{r}(t)$, we can turn back to $\vec{r}_1$ and $\vec{r}_2$ by using (*)

Planetary Motion - Kepler's problem;

If we consider the equation for the path of particle moving under the influence of central force $F_c \propto \frac{1}{r^2} \Rightarrow U(r) = -\frac{k}{r}$. Therefore

$$\theta(r) = \int \frac{\pm l/r^2 dr}{\sqrt{2\mu \left(E + \frac{k}{r} - \frac{l^2}{2\mu r^2} \right)}} + \text{constant.}$$

To calculate this integral it is useful to change the variable to $u \equiv \frac{l}{r}$

$$\frac{du}{dr} = -\frac{l}{r^2} \Rightarrow dr = -\frac{r^2}{l} du = -\frac{l r^2}{l^2} du = -l \frac{1}{u^2} \Rightarrow \boxed{dr = -\frac{l du}{u^2}}$$

$$\theta(u) = \int \frac{\frac{u^2}{l} \frac{l}{u^2} du}{\sqrt{2\mu \left(E + \frac{k}{l} u - \frac{1}{2\mu} u^2 \right)}} = \int \frac{du}{\sqrt{2\mu E + 2\mu \frac{k}{l} u - u^2}}$$

This integral looks like:

$$\int \frac{dx}{\sqrt{x^2 + bx + c}} = \int \frac{dx}{\sqrt{-(x - \frac{b}{2})^2 + \frac{b^2}{4} + c}} = \int \frac{dx}{\sqrt{P \left\{ 1 - \left[\frac{1}{P} \left(x - \frac{b}{2} \right) \right]^2 \right\}}}$$

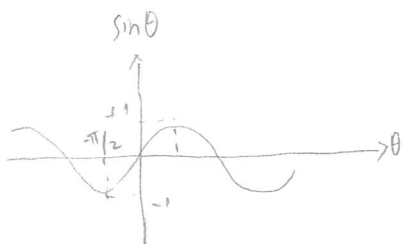
$$t = \frac{1}{\sqrt{P}} \left(x - \frac{b}{2} \right) \Rightarrow \frac{dt}{dx} = \frac{1}{\sqrt{P}} \Rightarrow dx = \sqrt{P} dt$$

$$= \int \frac{\sqrt{P} dt}{\sqrt{P} \sqrt{1-t^2}} = \int \frac{dt}{\sqrt{1-t^2}} = \arcsin(t) + C = \arcsin \left[\frac{1}{\sqrt{P}} \left(x - \frac{b}{2} \right) \right] + C = \arcsin \left[\frac{1}{\sqrt{\frac{b^2}{4} + c}} \left(x - \frac{b}{2} \right) \right] + C$$

• In our case, we have $c = 2\mu E$, $b = 2\mu k/l$

$$\theta(r) = \frac{1}{\sqrt{P}} \arcsin \frac{1}{\sqrt{\frac{4\mu^2 k^2}{4l^2} + 2\mu E}} \left(u - \frac{2\mu k}{2l} \right) + C \quad \text{or we can rearrange this equation as:}$$

$$\sin(\theta + C) = \frac{1}{\sqrt{\frac{\mu^2 k^2}{l^2} + 2\mu E}} \left(\frac{l}{r} - \frac{\mu k}{l} \right) = \frac{\frac{\mu k}{l} \left[\frac{l}{\mu k} \frac{l}{r} - 1 \right]}{\sqrt{\frac{\mu^2 k^2}{l^2} + 2\mu E}}$$



$$= \frac{\frac{l^2}{\mu k r} - 1}{\sqrt{\frac{l^2}{\mu^2 k^2} \left(\frac{\mu^2 k^2}{l^2} + 2\mu E \right)}} = \frac{\frac{l^2}{\mu k} \frac{1}{r} - 1}{\sqrt{1 + \frac{2El^2}{\mu k^2}}} \quad \begin{matrix} \text{positive} \\ \text{positive and constant} \end{matrix}$$

• When $r = r_{\min} \Rightarrow \frac{l^2}{\mu k} \frac{1}{r}$ is max $\sin(\theta + C)$ must be positive and max (+1) takes its max value.

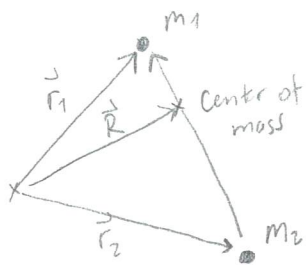
• So, just to get rid of C , we can choose the origin of θ such that when $\theta = 0$ $r = r_{\min} \Rightarrow$ This is equivalent of choosing $C = +\pi/2$ (we shift the $\sin \theta$ graph left by $\pi/2$)

$$\sin\left(\theta + \frac{\pi}{2}\right) = \cos \theta \rightarrow \text{finally}$$

$$\cos \theta = \frac{\frac{l^2}{\mu k} \frac{1}{r} - 1}{\sqrt{1 + \frac{2El^2}{\mu k^2}}}$$

Start with the definition of COM. Give the example of a bird flock.
What is this central-force problem really all about?

Central-force problem actually describes the motion of a system consisting two bodies affected by a force directed along the line connecting the centers of two bodies.



How to DESCRIBE THE SYSTEM:

- i) To describe the positions of m_1 and m_2 (and hence, to describe the system) we need to specify $\vec{r}_1$ and $\vec{r}_2$; 6 real quantities in total.
- ii) Alternatively we can consider $\vec{R}$ (showing the position of center of mass) and $\vec{r} = \vec{r}_1 - \vec{r}_2$ showing the relative positions of masses m_1 and m_2 .

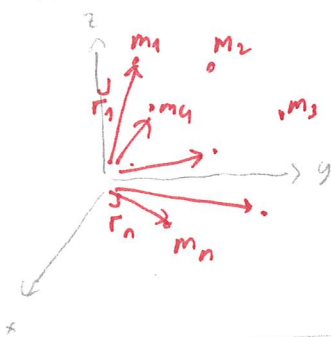
We assume 2 things:

1) No friction

2) The potential energy is only function of $r = |\vec{r}_1 - \vec{r}_2|$

★ Translation of the whole system is uninteresting from the point of view of the orbits of the particles, we can choose the origin of the coord. system to be particle's center of mass; $\vec{R} \equiv 0$

What is center of mass?



$$\vec{R}_{\text{COM}} = \frac{1}{\sum_i m_i} \sum_i m_i \vec{r}_i \rightarrow \text{Weighted sum of the radius vectors where weights are the masses.}$$

It is a useful quantity when it comes to describe the dynamics of a system of particles (system consisting of more than 1 particle)

For 2 body problem if we choose the COM as the origin:

$$\vec{R}_{\text{COM}} = m_1 \vec{r}_1 + m_2 \vec{r}_2 \equiv 0 \quad \text{when we combine this equation with } \vec{r} = \vec{r}_1 - \vec{r}_2$$

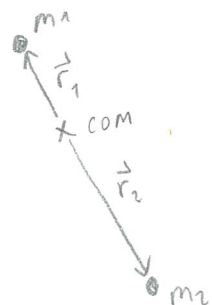
$$\vec{r}_1 = -\frac{m_2}{m_1} \vec{r}_2 \quad \text{and} \quad \vec{r}_2 = \vec{r}_1 - \vec{r}$$

$$\Rightarrow \vec{r}_1 = -\frac{m_2}{m_1} (\vec{r}_1 - \vec{r}) \Rightarrow \vec{r}_1 + \frac{m_2}{m_1} \vec{r}_1 = +\frac{m_2}{m_1} \vec{r} \Rightarrow \vec{r}_1 = \frac{m_2}{m_1} \vec{r} \left(1 + \frac{m_2}{m_1}\right)^{-1}$$

$$\Rightarrow \vec{r}_1 = \frac{m_2}{m_1} \left(\frac{m_1}{m_1 + m_2}\right) \vec{r} = \frac{m_2}{m_1 + m_2} \vec{r}$$

Also from

$$\vec{r}_2 = -\frac{m_1}{m_2} \vec{r}_1 \rightarrow \vec{r}_2 = \frac{m_1}{m_1 + m_2} \vec{r}$$



SOLUTION FOR TRAJECTORY FOR $U(r) = -k/r$

$$U(r) = -\frac{k}{r}$$

$$\theta(r) = \int \frac{\pm \frac{l}{r^2} dr}{\sqrt{2\mu \left(E + \frac{k}{r} - \frac{l^2}{2\mu r^2} \right)}} + \text{constant} \rightarrow \left(E - U - \frac{l^2}{2\mu r^2} \right)$$

$$U = \frac{k}{r} \rightarrow \frac{du}{dr} = -\frac{k}{r^2} \rightarrow dr = -\frac{r^2}{k} du \rightarrow \boxed{dr = -\frac{l}{u^2} du}$$

$$\theta(u) = \mp \int \frac{\frac{l}{r^2} \frac{r^2}{u^2} du}{\sqrt{2\mu \left(E + \frac{k}{u} - \frac{1}{2\mu} u^2 \right)}} + D = \mp \int \frac{du}{\sqrt{2\mu E + 2\mu \frac{k}{l} u - u^2}} + D$$

How to calculate this integral?

• The last integral looks like:

$$\int \frac{dx}{\sqrt{-x^2 + bx + c}} = \int \frac{dx}{\sqrt{-\left(x - \frac{b}{2}\right)^2 + \frac{b^2}{4} + c}} = \int \frac{dx}{\sqrt{p \left\{ 1 - \left[\frac{1}{\sqrt{p}} \left(x - \frac{b}{2} \right) \right]^2 \right\}}}$$

$t \rightarrow$ change variables.

$$t = \frac{1}{\sqrt{p}} \left(x - \frac{b}{2} \right) \rightarrow \frac{dt}{dx} = \frac{1}{\sqrt{p}} \rightarrow dx = \sqrt{p} dt$$

$$\int \frac{\sqrt{p} dt}{\sqrt{p} \sqrt{1-t^2}} = \int \frac{dt}{\sqrt{1-t^2}} = \arcsin(t) + D = \arcsin \left[\frac{1}{\sqrt{p}} \left(x - \frac{b}{2} \right) \right] + D$$

$$= \arcsin \left[\frac{1}{\sqrt{\frac{b^2}{4} + c}} \left(x - \frac{b}{2} \right) \right] + D$$

• In the original integral: $c \equiv 2\mu E$, $b \equiv 2\mu \frac{k}{l}$

$$\theta = 0 \quad r = r_{\min}$$

$$\theta(r) = \mp \arcsin \left[\frac{1}{\sqrt{\frac{\mu^2 k^2}{l^2} + 2\mu E}} \left(u - \frac{\mu k}{l} \right) \right] + D$$

they can be observed in D

$$(\theta + D) = \mp \arcsin \left[\frac{1}{\sqrt{\frac{\mu^2 k^2}{l^2} + 2\mu E}} \left(\frac{l}{r} - \frac{\mu k}{l} \right) \right]$$

$$\sin(\theta + D) = \frac{\frac{\mu k}{l} \left(\frac{l}{r} - \frac{\mu k}{l} \right)}{\sqrt{\frac{\mu^2 k^2}{l^2} + 2\mu E}} = \frac{\frac{l^2}{\mu k} \frac{1}{r} - 1}{\sqrt{1 + \frac{2E l^2}{\mu k^2}}}$$

positive

positive and constant

• When $r = r_{\min}$, $\frac{l^2}{\mu k r}$ is max
 $\Rightarrow \sin(\theta + D)$ must be positive and max. (+1)

$$\xi = \sqrt{1 + \frac{2El^2}{\mu k^2}}$$

$$E = 0 \Rightarrow \xi = 1 : \text{Parabola}$$

$$E > 0 \Rightarrow \xi > 1 : \text{Hyperbola}$$

$$V_{\min} < E < 0 \Rightarrow 0 < \xi < 1 : \text{Ellipse}$$

$$E = V_{\min} \Rightarrow \xi = 0 : \text{Circle.}$$

$$\text{For } \xi = 0 \quad \frac{2El^2}{\mu k^2} = -1$$

$$\frac{E}{\frac{\mu k^2}{2l^2}} = -1$$

$$\boxed{E = -\frac{\mu k^2}{2l^2}} = V_{\min.}$$

$$\frac{\mu k^2}{2l^2} > 0 \text{ by definition.}$$

$$V_{\min} = -\frac{\mu k^2}{2l^2}$$

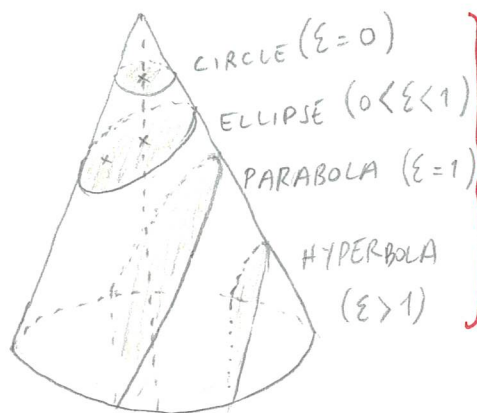
$$\sqrt{1 + \frac{E}{V_{\min}}}$$

Let us define the constants $\alpha \equiv l^2/\mu k$ and $\epsilon \equiv \sqrt{1 + \frac{2El^2}{\mu k^2}}$

Then $\cos\theta$ equation can be written as:

$$\cos\theta = \frac{\frac{\alpha}{r} - 1}{\epsilon}$$

$$\rightarrow \frac{\alpha}{r} = 1 + \epsilon \cos\theta \rightarrow \text{Equation of a conic section with one focus at the origin}$$



defining the angle of the plane.

ϵ : eccentricity
 2α : latus rectum.

intersection of a plane and cone.

Focus is the center of force.

★ As we have chosen previously r_{min} is obtained when $\theta=0$
 $\Rightarrow \cos\theta \rightarrow \max$ and $\alpha/r \rightarrow \max$
 $\Rightarrow r_{min}$.

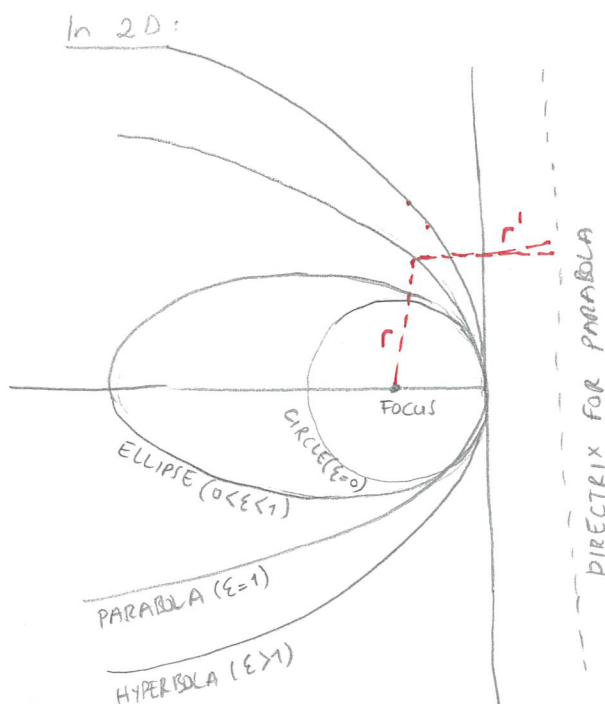
- Periater: $\theta=0 \rightarrow r_{min}$
- Apoater: $\rightarrow r_{max}$

→ For the motion about the sun
perihelion and aphelion
→ For the motion about the earth
perigee and apogee

Eccentricity and total energy

$$\epsilon = \sqrt{1 + \frac{2El^2}{\mu k^2}} \rightarrow \frac{1}{|V_{min}|}$$

- $\epsilon > 1$, $E > 0$: Hyperbola
- $\epsilon = 1$, $E = 0$: Parabola
- $0 < \epsilon < 1$, $V_{min} < E < 0$: Ellipse.
- $\epsilon = 0$, $E = V_{min}$: Circle



$$\epsilon = \sqrt{1 - \frac{E}{V_{min}}}$$

$$E = V_{min}, \epsilon = \sqrt{1 - (-1)}$$

Exercise:

Determining V_{min} :

$$V(r) = -\frac{k}{r} + \frac{l^2}{2\mu r^2}$$

$$\frac{dV}{dr} = \frac{k}{r^2} - \frac{l^2}{\mu r^3} = 0 \rightarrow \frac{k}{r} = \frac{l^2}{\mu r^2}$$

$$r = \frac{l^2}{\mu k}$$

$$V_{min} = -\frac{k\mu k}{l^2} + \frac{l^2}{2\mu \left(\frac{l^2}{\mu k}\right)^2}$$

$$= -\frac{k^2\mu}{l^2} + \frac{k^2\mu}{2l^2} = -\frac{2k^2\mu}{2l^2} + \frac{k^2\mu}{2l^2} = -\frac{k^2\mu}{2l^2}$$

when $E = V_{min}$:

$$\epsilon = \sqrt{1 - \frac{2l^2}{\mu k^2} \frac{k^2\mu}{2l^2}} = 0 \Rightarrow \epsilon = \sqrt{1 - E/V_{min}}$$

In terms of the orbit equation:

- $\epsilon = 0$: $r = \alpha = \frac{\ell^2}{\mu k}$: The radius of the circle α depends on ℓ^2 , μ and k

what is k ?

$$U(r) = -\frac{k}{r} \quad F = -\frac{dU}{dr} = -\frac{k}{r^2} = \frac{G m_1 m_2}{r^2}$$

$k = G m_1 m_2$
 Grav. const. $\downarrow$ masses of the objects

- $0 < \epsilon < 1$: Ellipse : DESCRIBES PLANETARY MOTION

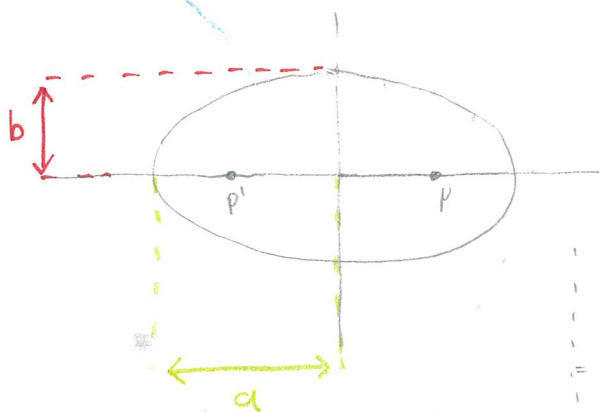
$$r_{\min} : \theta = 0 \rightarrow \frac{\alpha}{r_{\min}} = 1 + \epsilon \rightarrow r_{\min} = \frac{\alpha}{1 + \epsilon}$$

$$r_{\max} : \theta = \pi \rightarrow \frac{\alpha}{r_{\max}} = 1 - \epsilon \rightarrow r_{\max} = \frac{\alpha}{1 - \epsilon}$$

$$r_{\min} + r_{\max} = \frac{\alpha(1 - \epsilon) + \alpha(1 + \epsilon)}{1 - \epsilon^2}$$

$$= \frac{\alpha[1 - \epsilon + 1 + \epsilon]}{1 - \epsilon^2}$$

$$= \frac{2\alpha}{1 - \epsilon^2} \quad \text{length of major axis}$$



Semi major axis

$$a = \frac{\alpha}{1 - \epsilon^2} = \frac{\ell^2}{\mu k} \frac{1}{1 - 1 + \frac{2E\ell^2}{\mu k^2}} = \frac{k}{2E}$$

$$b = \frac{\alpha}{\sqrt{1 - \epsilon^2}} = \frac{\ell^2}{\mu k} \frac{1}{\sqrt{1 - 1 + \frac{2E\ell^2}{\mu k^2}}} = \frac{\ell^2}{\sqrt{\mu k^2 \frac{2E\ell^2}{\mu k^2}}} = \frac{\ell^2}{\sqrt{2\mu |E|}}$$

Semi minor axis.

H.W.
 • To find the length of the minor axis:

Find the turning point of $r \sin \theta$.

$$r = \frac{\alpha}{1 + \epsilon \cos \theta} \Rightarrow \frac{d}{d\theta} \left[\frac{\alpha \sin \theta}{1 + \epsilon \cos \theta} \right]$$

$$\frac{d}{d\theta} \left\{ [\alpha \sin \theta] [1 + \epsilon \cos \theta]^{-1} \right\} = \alpha \cos \theta [1 + \epsilon \cos \theta]^{-1} + (-1) [1 + \epsilon \cos \theta]^{-2} \epsilon \sin \theta \sin \theta = 0$$

$$\Rightarrow \cos \theta = -\frac{\epsilon \sin^2 \theta}{1 + \epsilon \cos \theta} \Rightarrow \epsilon \cos^2 \theta + \cos \theta = -\epsilon \sin^2 \theta$$

$$\Rightarrow \epsilon (\cos^2 \theta + \sin^2 \theta) + \cos \theta = 0$$

$$\Rightarrow \cos \theta = -\epsilon \rightarrow \theta = \arccos(-\epsilon)$$

$$\text{At this angle } \frac{\alpha}{r} = 1 + \epsilon(-\epsilon) \Rightarrow \frac{\alpha}{r} = 1 - \epsilon^2$$

$$\Rightarrow r = \frac{\alpha}{1 - \epsilon^2}$$

• Half minor axis:

$$\Rightarrow r \sin \theta = \frac{\alpha}{1 - \epsilon^2} \sin \{ \arccos(-\epsilon) \} = \frac{\alpha}{1 - \epsilon^2} \sqrt{1 - \epsilon^2}$$

$$= \frac{\alpha}{\sqrt{1 - \epsilon^2}}$$

The Period of the Elliptic Motion:

$$\frac{dA}{dt} = \frac{1}{2\mu} \rightarrow dt = \frac{2\mu}{\ell} dA$$

Entire area of an ellipse is swept in one complete period τ :

$$\int_0^\tau dt = \frac{2\mu}{\ell} \int_0^A dA \rightarrow \tau = \frac{2\mu}{\ell} A \quad \text{we know that area of an ellipse is } A = \pi ab$$

$$\tau = \frac{2\mu}{\ell} \pi ab = \frac{2\mu}{\ell} \pi \frac{k}{2|E|} \frac{\ell^2}{\sqrt{2\mu |E|}} = \frac{\mu \pi k}{\sqrt{2\mu |E|^3}} = \pi k \sqrt{\frac{\mu}{2}} |E|^{-3/2}$$

Alternatively:

$$a = \frac{\alpha}{1 - \epsilon^2} \quad b^2 = \frac{\alpha^2}{1 - \epsilon^2} \rightarrow \frac{a}{b} = \frac{\alpha^2}{b^2} \rightarrow b = \sqrt{a} \Rightarrow \tau = \frac{2\mu \pi a \sqrt{a}}{\ell} \Rightarrow \tau^2 = \frac{4\mu^2 \pi^2 a^2 a}{\ell^2}$$

$$\Rightarrow \tau^2 = \frac{4\mu^2 \pi^2 \ell^2}{\mu k} a^3 = 4\pi^2 \frac{\mu}{k} a^3$$

$$\tau^2 = 4\pi^2 \frac{\mu}{k} a^3$$

IMPORTANT RESULT: $\tau^2 = 4\pi^3 \frac{\mu}{k} a^3$: square of the period is proportional to the cube of semimajor axis of the elliptic orbit, known as **KEPLER'S THIRD LAW!**

- ★ This result is valid for equivalent one-body problem in a central force field.
- ★ From the very precise experimental data obtained by Tycho Brahe, Kepler concluded that $\tau^2 \propto a^3$ with the same proportionality constant for all planets. In this sense, his result is only approximately correct because reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is different for each planet.

Since $\vec{F}(r) = - \frac{G m_1 m_2}{r^2} = - \frac{k}{r^2}$ $k \equiv G m_1 m_2$

→ k changes acc. to the total mass of the 2 body problem.

$$\Rightarrow \tau^2 = 4\pi^3 \frac{m_1 m_2}{m_1 + m_2} \frac{a^3}{G m_1 m_2} = \frac{4\pi^3 a^3}{G(m_1 + m_2)} \approx \frac{4\pi^3 a^3}{G m_2} \quad m_1 \ll m_2$$

- ⚠ Kepler's statement is correct only if the mass m_1 of a planet can be neglected with respect to the mass m_2 of the sun.

Kepler's laws of planetary motion:

- i) Planets move in elliptical orbits about the sun with sun at one focus.
- ii) The area per unit time swept out by a radius vector from sun to a planet is constant.
- iii) The square of a planet's period is proportional to the cube of the major axis of the planet's orbit.

Stability of circular orbits (Optional)

- Circular orbit if $E = V_{\min}$
- A circular orbit is allowed for any attractive potential because the attractive force can always be made to just balance the "centrifugal force" by proper choice of radial velocity.
- Although circular orbits are always possible in a central attractive force field, such orbits are not necessarily stable.
- A circular orbit $r=s$ exists if $\dot{r}|_{r=s} = 0$ for all t ; this is possible if $\left. \frac{\partial V}{\partial r} \right|_{r=s} = 0$ but only if the effective potential has a true minimum does stability result. All other equilibrium ~~are~~ circular orbits are unstable.

$$F(r) = -\mu g(r) = -\frac{\partial U}{\partial r}$$

$$F(r) = \mu(\ddot{r} - r\dot{\theta}^2)$$

$$\frac{F(r)}{\mu} = \ddot{r} - r\dot{\theta}^2 = -g(r)$$

$$l = \mu r^2 \dot{\theta} \rightarrow \dot{\theta} = \frac{l}{\mu r^2}$$

$$-g(r) = \ddot{r} - \cancel{r} \frac{l^2}{\mu^2 r^4} = \ddot{r} - \frac{l^2}{\mu^2 r^3}$$

- Assume that we have a particle to be initially in a circular orbit with radius s and apply a perturbation of the form $r \rightarrow s+x$, where x is small.
 $\ddot{r} \rightarrow \ddot{x}$ since s is constant. Then,

$$\ddot{r} - \frac{l^2}{\mu^2 r^3} = -g(r) \rightarrow \ddot{x} - \frac{l^2}{\mu^2 (s+x)^3} = -g(s+x)$$

$$\ddot{x} - \frac{l^2}{\mu^2 s^3 \left(1 + \frac{x}{s}\right)^3} = -g(s+x)$$

- According to our assumption $\frac{x}{s} \ll 1$,

$$\left(1 + \frac{x}{s}\right)^{-3} = 1 - 3\frac{x}{s} + \dots$$

- And also $g(s+x)$ can be expanded in Taylor series about point $r=s$

Examples for tutorial hour (13.11.2025) (About central-force motion)

1) Show that $\langle T \rangle = -\frac{1}{2} \langle U \rangle$ for the particles under gravitational attraction.

Let us define a ^{new} quantity: $S \equiv \sum_i \vec{p}_i \cdot \vec{r}_i$

$$\frac{dS}{dt} = \sum_i \left(\vec{p}_i \cdot \dot{\vec{r}}_i + \dot{\vec{p}}_i \cdot \vec{r}_i \right) \Rightarrow \left\langle \frac{dS}{dt} \right\rangle = \frac{1}{2} \int_0^2 \frac{dS}{dt} dt = \frac{S(2) - S(0)}{2}$$

★ If system's motion is periodic and if τ is some integer multiple of the period then $S(\tau) = S(0)$ and $\langle \dot{S} \rangle$ vanishes.

★ If the system is not periodic but it is bound (S has a finite value) we can always keep τ as long as possible to make $\langle \dot{S} \rangle$ as small as possible

$$\lim_{\tau \rightarrow \infty} \frac{S(\tau) - S(0)}{2} = \lim_{\tau \rightarrow \infty} \langle \dot{S} \rangle = 0 \quad \text{for a bound system.}$$

$$\langle \dot{S} \rangle = \left\langle \sum_i \vec{p}_i \cdot \dot{\vec{r}}_i \right\rangle + \left\langle \sum_i \dot{\vec{p}}_i \cdot \vec{r}_i \right\rangle = 0$$

$$\Rightarrow \left\langle \sum_i \vec{p}_i \cdot \dot{\vec{r}}_i \right\rangle = - \left\langle \sum_i \dot{\vec{p}}_i \cdot \vec{r}_i \right\rangle$$

$$\left\langle \sum_i \underbrace{m_i \vec{v}_i \cdot \vec{v}_i}_{2T_i} \right\rangle$$

$$\left\langle 2 \sum_i T_i \right\rangle$$

$$\left\langle 2 \sum_i T_i \right\rangle = - \left\langle \sum_i \vec{F}_i \cdot \vec{r}_i \right\rangle$$

$$\Rightarrow \langle T \rangle = \left(-\frac{1}{2} \left\langle \sum_i \vec{F}_i \cdot \vec{r}_i \right\rangle \right) \rightarrow \text{Virial (coined by Clausius)}$$

Virial theorem: The average kinetic energy of a system of particles is equal to its virial.

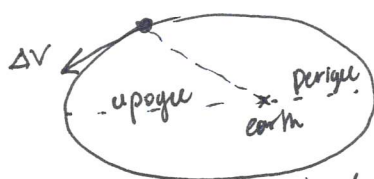
- 3) An Earth satellite moves in an elliptical orbit with a period τ , eccentricity e , and semi-major axis a . Show that max radial velocity is $\frac{2\pi a e}{\tau \sqrt{1-e^2}}$. (Given: $\tau = \frac{2\pi}{\omega} \pi a b = \frac{2\pi \mu a^3}{\ell \sqrt{1-e^2}}$ $\rightarrow \frac{\ell}{a} = \frac{2\pi \mu a}{\tau \sqrt{1-e^2}}$)

time derivative $\downarrow$ start with $\frac{a}{r} = 1 + e \cos \theta$

$$\therefore \frac{a}{r^2} \dot{r} = -\dot{\theta} e \sin \theta \Rightarrow \frac{\dot{r}}{r^2} = \frac{\dot{\theta} e}{a} \sin \theta \Rightarrow \frac{\dot{r}}{r^2} = \frac{\ell}{\mu r^2} \frac{e}{a} \sin \theta$$

$$\dot{r} = \frac{\ell e}{\mu a} \sin \theta \quad | \dot{r} |_{\max} = \frac{\ell e}{\mu a} = \frac{e}{\mu} \frac{\ell}{a} = \frac{e}{\mu} \frac{2\pi \mu a}{\tau \sqrt{1-e^2}} = \frac{2\pi a e}{\tau \sqrt{1-e^2}}$$

- 4) What is the most efficient way to change the energy of an elliptical orbit for a single short engine thrust?



First consider a velocity kick ΔV applied along the direction of travel at an arbitrary place in the orbit. We seek the optimum location to apply this kick;

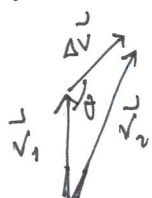
initial energy $\rightarrow E_1 = \frac{1}{2} m v^2 - \frac{GMm}{r}$

final energy $\rightarrow E_2 = \frac{1}{2} m (v + \Delta v)^2 - \frac{GMm}{r}$

Energy gain $E_2 - E_1 = \frac{1}{2} m (v^2 + 2v\Delta v + \Delta v^2 - v^2) = \frac{1}{2} m (2v\Delta v + \Delta v^2)$

For a given Δv $E_2 - E_1$ is maximized when v is max, at perigee.

Optimal direction of $\Delta \vec{v}$: If $\Delta \vec{v}$ is applied at perigee in an arbitrary direction the total velocity vector after the kick is



New K.E

$$\vec{V}_2 = \vec{V}_1 + \Delta \vec{V}$$

$$\frac{1}{2} m v_2^2 = \frac{1}{2} m (v_1^2 + \Delta v^2 + 2v_1 \Delta v \cos \theta)$$

to maximize v_2^2 and thus the orbital energy, we $\cos \theta$ must be maximised

$$\Rightarrow \theta = \pi \text{ when } \vec{V}_1 \parallel \Delta \vec{V}$$

Apply Δv along the direction of motion at perigee.